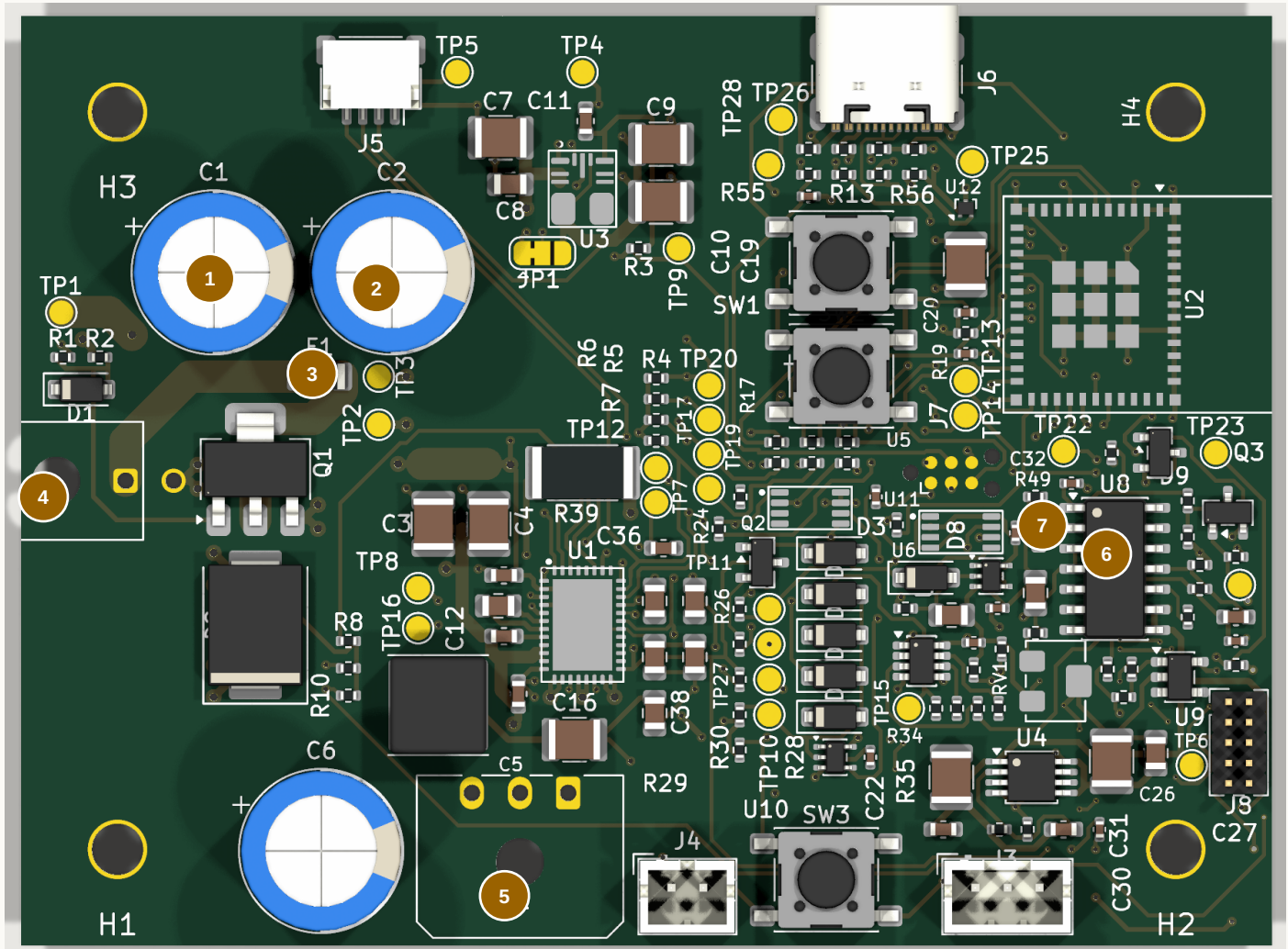


1A PCB-01 Population Map

FIELD GUIDE

TOP / COMPONENT SIDE - KiCad target appearance



1 C1
5 J2

2 C2
6 U8

3 F1
7 C34

4 J1

SOLDER ORDER: U8 -> C34 -> J1/J2 -> C1/C2 -> INSPECT -> F1 LAST

1 + 2 C1 / C2

EEU-FR1H471, 470 uF 50 V. Positive lead to board +; negative stripe to pin 2 / PGND. Seat square without crushing bung.

3 F1 - INSTALL LAST

Low-profile tinned copper link across the 1206 pads. Near-zero ohms after soldering. External wall-side 3 A fuse remains mandatory.

4 + 5 CONNECTORS

J1 43045-0200; J2 43650-0300. Larger chisel tip. Tack one electrical pin, square and fully seat housing, then finish. Locating pegs are not soldered.

6 U8 / 7 C34

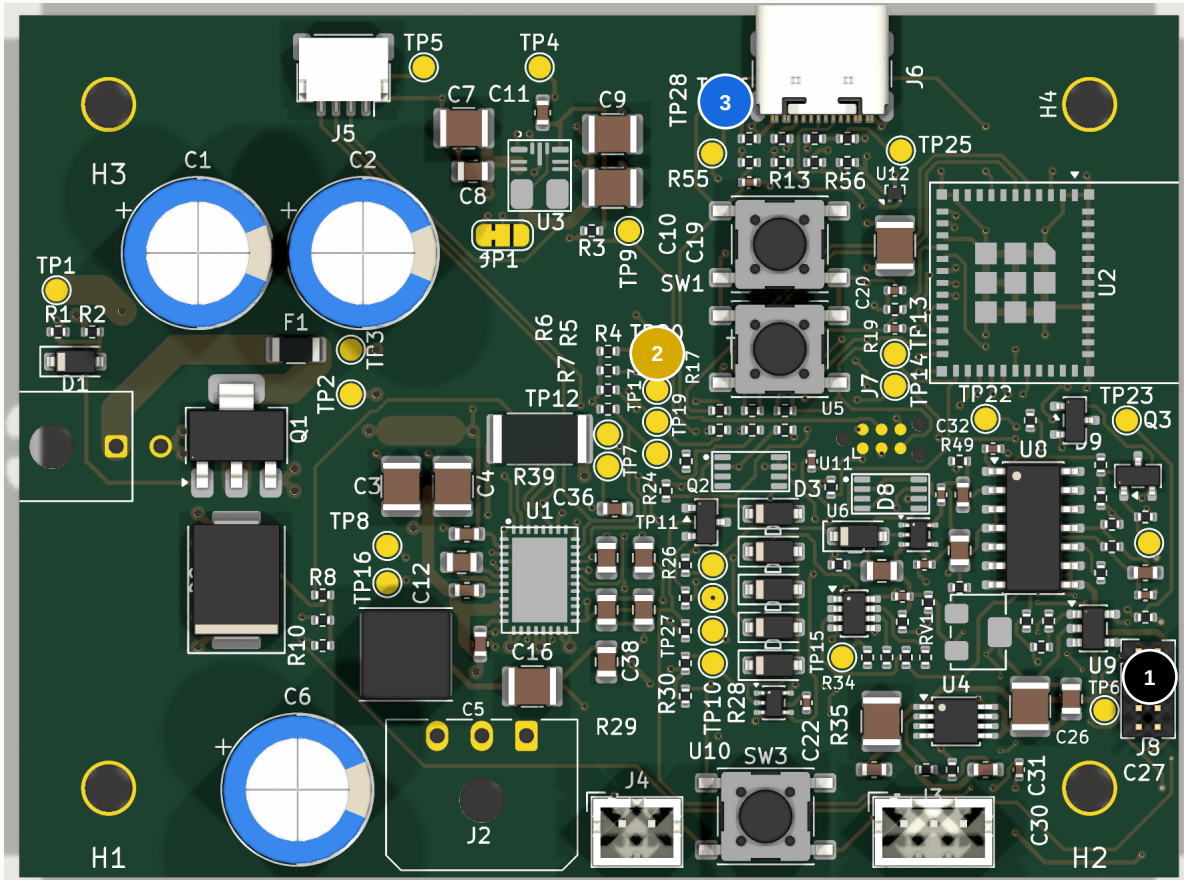
U8 LM2907M/NOPB: notch/dimple to pin-1 marker, tack pin 1 then diagonal pin 8, flux and drag-solder. C34 C1206C104K3GACTU: 100 nF COG 25 V +/-10%, spanning the smaller 0603 site, not shorting it.

PCB-01: USE THE IRON, NOT BROAD HOT AIR. Then clean and dry; inspect under the microscope; verify no hard short from RAW24, VM24, 3V3, or 12V_TACH to ground; photograph the board.

1B Removable Tuning Leads

FIELD GUIDE

INSTALL ON THE BENCH BEFORE THE BOARD GOES OVERHEAD



1 J8.9 SOX - BLACK

2 TP20 FG - YELLOW

3 TP26 AGND - BLUE

FIXED TUNING SETUP - DO NOT MOVE THE SCOPE

Physical leads: J8.9 SOX black, TP20 FG yellow, TP26 AGND blue. OWON CH1 uses black, CH2 uses yellow, and both common channel grounds use blue. Keep this hookup unchanged through unloaded and loaded tuning. Firmware already records SPEED, NFAULT, and DRVOFF.

VM24: ONE-TIME BEFORE TUNING

If the required bus capture is still open, complete it before fixing the OWON to these leads: x10 tip directly on TP2, spring ground on TP3. Do not substitute VM24 into CH2 during tuning.

FIT, INSPECT, REMOVE AFTER FINAL TUNING

All sources removed and bulk capacitors discharged. Solder once, route insulated wire flat, add Kapton strain relief away from each joint, inspect under magnification, then continuity-check every lead to its named net and for adjacent-pad shorts. Cap every free end before power. TP4 is also usable AGND; never wire motor phases, switch nodes, or USB data pads. Remove the pigtailed after the complete exposed and housed tuning campaign, then clean and inspect.

OWON CHANNEL GROUNDS ARE COMMON. Never put one ground on AGND and the other on PGND in the same setup.